

# SAMUEL EHRLICH

[sehrlich98@gmail.com](mailto:sehrlich98@gmail.com) [X](#) [LinkedIn](#) [Github](#)

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## PROFESSIONAL EXPERIENCE

Driveline Baseball, Kent, WA

Jan 2025 - Present

### ***Baseball Operations Analyst - Data Science***

- Repeat Researcher and presenter at 2x SABR and 1x Saberseminar conference on novel batting and pitching studies
  - Biomechanics and Command Interaction study awarded the [SABR Mike Marshall's Biomechanics Research Award](#)
  - [Swing Path Optimization](#) identifying the optimal Power vs Contact tradeoff of a batter's swing using a pareto frontier
  - [Pitch physics study](#) on seam shifted wake to maximize efficiency of pitches utilizing different seam orientations
- Streamlined advance scouting reports and AI integration to provide matchup insights to pro level clients
- Developed industry leading pitch models using a component based approach to isolate parts of the pitch into specific contributions
  - Stacking up against FanGraphs with an  $R^2$  of over 0.75 across various pitch types and event specific qualifications
- Refined skills and tools in Claude for our less technical employees to automate their workflows and save time on repeatable tasks
- Built an end-to-end CV pipeline to [detect baseball seams](#) from Edgertonic footage using CVAT/Roboflow for data annotation, SAM3 fine-tuned for segmentation, and optimizing the model using autoresearch and other techniques to improve mAP50-95
- Built in-house BI Dashboard "Samcast" for our trainers saving \$100k annually utilizing MongoDB, Clickhouse, FastAPI, and React
  - Collaborated with trainers to own and maintain a product used company-wide daily and shared across social media
  - Used Karpathy's Wiki LLM framework with gbrain to efficiently route context during development and debugging

University of Missouri Baseball, Columbia, MO

Aug 2022 - May 2023

### ***Data Analyst***

- Built numerous machine learning models to identify multiclass pitch types using XGBoost, KNN, and clustering methods
- Developed a custom Elo system to predict win probability of games and strength of teams at different points in the season
- Designed interactive dashboards using R Shiny to assist in driving decision making for our coaching staff on and off the field

## EDUCATION

University of Missouri - Columbia, Missouri

Aug 2021 - May 2023

### **Master of Science in Data Science and Analytics**

Major in High Performance Computing - GPA 3.93

- **Studies:** Statistics, Visualization, Web Scraping, Data Mining, Database Design, Machine Learning, Cloud Computing

### **Bachelor of Science in Parks, Recreation and Sport**

Aug 2017 - May 2020

Major in Sport Management, Minor in Hospitality Management - GPA: 3.66 (Cum Laude, Dean's List)

## PROJECTS

Contributed to multiple Driveline research studies and wrote about them for the Driveline Blog ([Bat Path](#), [Force Plates](#), [Command Study](#))

Developed [Twitter](#) account to share research findings and engage with the analytics community through content creation and discussions

- Built a community of over 1000 followers with posts featured on [Pitching Ninja](#) and [MLB](#) with engagement over 1M

"[NCAA Tournament Sim](#)" Elo ranking model for NCAA D1 Baseball teams based on historical performances using data from 2021-2024

- Successfully predicted 3 of the final 8 teams that made it to Omaha in the 2025 Men's College World Series
- Generated NCAA tournament championship probability forecasts for all teams through Monte Carlo simulation of 20,000 brackets

"[Stuff Model](#)" Pitching grade model that applies run values to a pitch solely on the movement profile, agnostic of location

- Data cleaning, EDA using domain knowledge, research, clustering methods, and sampling techniques for imbalance classes
- Utilized XGBoosting paired with Optuna for hyperparameter tuning to generate predictions that best minimized the MSE

## TECHNICAL SKILLS

**Languages and Software:** SQL (PostgreSQL, MySQL, NoSQL, MongoDB, Clickhouse), R (Rstudio, Shiny, Tidyverse, ggplot2, xgboost), Python (Pandas, Numpy, Matplotlib, Seaborn, Plotly, scikit-learn, BS4, cv2, ultralytics, mediapipe, FastAPI, optuna), Cloud Computing (GCP, AWS), Applied AI (Claude, OpenAI, Gemini, MoonDream, SAM3, autoresearch) React, Github, Hugging Face, CVAT, Roboflow

**Skills & Certs:** [Claude AI Fluency Certification](#), Applied Machine Learning, Advanced Visualization, Parameter Tuning, Data Engineering, CV, Bayesian and statistical modeling, Computer Vision (Object Detection, Segmentation, Image Classification, Visual Tracking), Hyperparameter Optimization